MTH 250.2
Elementary Differential Equations

Lecture Notes

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Contents

1 Introduction to Differential Equations

A function is a class of functions, i.e. "a function" is not a constant.

$y = 6x^2 + 5x + 7 \quad \cdots (1)$ $\dfrac{dy}{dx} = 12x + 5 \quad \cdots (11)$ (traditional algebraic equation)

In an algebraic equation, the solution to y is always a "constant," or set of constants (values).

> When you are given a differential equation with an initial condition, you **must** have solved.

An equation that contains a derivative $\left(\dfrac{dy}{dx}\right)$ is called a **Differential Equation**. It contains at least one derivative of the unknown function.

> A differential equation with an Initial Condition is called an **Initial Value Problem (IVP)**.
> "Boundary Value Problems" ...

To solve the differential equation, you reverse using integration:

$$\int \frac{dy}{dx}\, dx = \int (12x + 5)\, dx$$

$$\int dy = \frac{12x^2}{2} + 5x + C \qquad \text{(arbitrary constant)}$$

$$\therefore\ y = 6x^2 + 5x + C \quad \text{(General Solution)}$$

To determine the value of "Initial Condition": $y(0) = 1,\ y(0) = 4$.

Soln. Solution to a differential equation is when you introduce an initial condition to get the value of C.

> **NB:** Particular solution occurs with the substitution of initial condition to get the value of "C". After returning the C value into the General Solution, you get a Particular Solution.

1.1 Order and Degree

1) $y' + 2y - 3 = 0$ (1^{st} Order)
2) $y'' + 2y' + 3y = 0$
3) $\dfrac{2d^2y}{dt^2} - \dfrac{3dy}{dt} + t = 5$

Differentiating a dependent variable in the equation, differentiating with respect to a most independent variable.

eqn:

$$y = 2x^2 + 5x, \qquad \frac{dy}{dx} = 4x + 5$$

D is a differential function because it contains an unknown function (y) and contains one derivative (dy/dx) of that function.

Unknown: It is also a differential function, i.e. it depends on independent variables, or depends only on the independent variable.

i) Is a differential function because it contains an unknown function (y) and contains at least one derivative (dy/dx) of that unknown.

ii) Contains only one independent variable.

iii) Contains more than one independent variable with respect to more than one derivative.

Ordinary / Differential Equation, contains only one variable (t, c) etc. (CODE)

$$1)\ y' + 2y - 3 = 0 \Rightarrow \text{1st Order}$$

$$2)\ y'' + 3y'\Big|_{d\frac{d^2y}{dt^2}+3\frac{dy}{dt}+3y} = 0$$

$$3)\ 2\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + t = 5 \qquad \Rightarrow \qquad \frac{2d^2y}{dt^2} - \frac{3dy}{dt} + t = 5$$

Differentiating a dependent variable with respect to an independent variable.

eqn. $y = 2x^2 + 5x \Rightarrow y' = 4x + 5$

D is a differential function because it contains an unknown function (y) and its derivative.

D is also a differential function because it contains an unknown function (y) and depends on independent variable(s).

PDE (Partial Differential Equation) because it contains 2 derivatives: the ones with 2 derivatives, the one with the highest order used in naming the function.

$$\text{i.e. } y = x^2z^2 + 3z, \qquad \frac{\partial y}{\partial x} = 2xz^2 + 3z$$

You'll use one of the independent variable as a constant.

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eqn.

$$2y''' + y'' - 12y' = 5t$$

$$\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} = 2$$

1.2 Order, Degree and Type Classification

Characteristics of ODE: the dependent variable and all its derivatives are a function of powers.

Order → Highest derivative that appears in the equation.

Degree → Power of the highest order derivative in the equation, after all the coefficients $\{h(x)\}$ are rational (contains more than one independent variable).

$$\frac{\partial^2 y}{\partial z^2} + 9\frac{\partial^2 y}{\partial z^2} = 0$$

$$\left(\frac{d^3y}{dx^3}\right) + \cot x\frac{d^2y}{dx^2} + 7xy = 0$$

Eqn.	Type of equation	Order	Degree
i	Ordinary	1	1
ii	Ordinary	3	$\neq 1/2$
iii	Ordinary	4	6
iv	Partial	2	1
v	Ordinary	3	1
vi	Ordinary	1	1
vii	Ordinary	4	1
viii	Partial	2	1
ix	Ordinary	3	1

Classes of D.E

- Linear D.E (L.D.E)
- Non-Linear D.E

Note: If any condition is not satisfied, then the D.E is said to be Non-linear D.E.

Characteristics of a Non-L.D.E

- $y'' \cdot y'''$ (When two or more derivatives multiply each other)
- $\left(\dfrac{d^2y}{dx^2}\right)$ or $y' \cdot y''$ (When y or y' has any power aside 1) exponential/trig functions

Sample equations:

$$y'' \text{ or } y^4, \quad \left(\frac{d^2y}{dx^2}\right)^3, \ (y')^5 \text{ or } y^2$$
$$xy'' + xy^2y + 2y = e^{-x} \ \ \text{(Non-L)}$$
$$y'' + xy^2y = 0 \ \ \text{(Non-L)}$$
$$xy'' - 2y + y = 0 \ \ \text{(Non-L)}$$

If the derivative of y or y' has any power aside 1.

For a D.E to be classified as Linear, all of the following conditions must hold:

- Dependent variable (y) and its derivatives are either a constant, or a function of the independent variable (x).
- Coefficient (constant, or coefficient function). The coefficient function should not multiply the dependent variable or its own derivative(s).
- An independent variable (x) should multiply a constant, or an independent variable.

NB: If any one of these two is not satisfied, then the D.E is said to be Non-linear.

Non-linear D.E is said to be Non-linear if $\left|\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right|$, then it should be Non-linear.

Characteristics of a Non-linear D.E

- $y'y''$ (When two or more derivatives are of exponential and product of derivative $(y' \cdot y''/x)$)
- $y' \cdot y''$ (exponential)

Eqn.	Classify into L.D.E or Non L.D.E
i)	$y'' + xy' + x^2y = e^{-x}$ (Non-L.)
ii)	$y'' - 3y' + xy = 0$ (Non-L.)
iii)	$xy'' - 2y' + y = 0$ (Non-L.)
iv)	$y''' + y = 0$ (L.D.E)
v)	$x'' + y'' + xy' + Iy = 0$
vi)	$\sin y \cdot \tan y$, $\sin(x, y)$ – Trigonometric ratios must only carry the independent variable for
vii)	$y'' + (y')^2 + 2y''' = 0$
viii)	$x'' + x = 0$ (L.D.E)
ix)	$xx' \cdot \sin(x)\, y'' + \ldots$

If it be linear. But if it (carries) independent variable for its own derivatives, then it becomes Non-L.D.E.

1.3 Methods of Variable Separation

$$e.g.\ \frac{dy}{dx} = 2\sin y$$

$$\int dy = \int (2x^2 - 3)\, dx \qquad \text{direct integration}$$

$$y = \frac{2x^3}{3} - 3x + c$$

A D.E = differentiated function, but not yet solved.

Any eqn of this form $\frac{dy}{dx} = F(x) \cdot G(y)$ is a separable variable D.E, when x and y (functions) are Products (Multiplied), while $F(x) \cdot G(y)$ is a separable variable equation, is a function of this form:

$$\frac{dy}{dx} = f(x) \cdot g(y)$$

2 Forming a D.E. from the Primitive

$y = ke^{3x}$ (Function)

Eliminate the constant "k" in the general solution.

Solution to a D.E: A Natural / of Order $\dfrac{d^n y}{dx^n} = f(x, y, \ldots, \frac{d^{n-1}y}{dx^{n-1}})$

Methods of finding one solution

(a) First, since it only has 1 constant, let Order, since it only has 1 constant, its solution to the D.E = find y', sub y' for $\dfrac{dy}{dx}$ and investigate that $y = ke^{3x}$ satisfies the D.E.

(b) Incorporate "y" into the derivative of $y' = 3y - \cdots (1)$

$$y = ke^{3x}$$

$$\frac{dy}{dx} = 3ke^{3x} \qquad \cdots (i)$$

$$\frac{dy}{dx} = 3ke^{3x} \qquad \cdots (ii)$$

Since $y = ke^{3x}$, subs (i) into (ii):

$$y' = 3ke^{3x} \qquad \cdots (iii)$$

$$\therefore \frac{dy}{dx} = 3y \qquad \cdots (i)$$

$$y' = 3y \qquad \cdots (i)$$

Thus by substitution, the arbitrary constant has been eliminated.

$$\therefore \; y' - 3y = 0 \qquad \text{D.E}$$

as x is moving, dividing y to one side, as $y' = 3y$; if P its multiplying, while $y' = f(x) \cdot g(y)$

$$\frac{dy}{dy} = 3\,dx$$

$$\therefore \int \frac{dy}{dy} = \int 3\,dx$$

$3 \ln |y| = 3x + K_0 \Rightarrow y = ke^{3x}$ is a solution to the D.E.

2.1 General Methods of Finding One Solution

Given a de-primitive $y = ke^{3x}$ (Function of Nature, Order in C, $C_1, C_2, \ldots$), eliminate the constant "k" in the general solution, then investigate that it should solve to the D.E of order, since it only has 1 constant.

$$\frac{dy}{P(x)} \quad \text{— find } y',\ \text{sub } y' \text{ for both P and } \frac{dy}{dx}$$

> Note: Exponential of $B_3(t)$ values, cannot be 0.

$$a(m^2 e^{mx}) + b(m\, e^{mx}) + c(e^{mx}) = 0 \qquad \text{Sub (iv)}$$

$$e^{mx}(am^2 + bm + c) = 0$$

$$a \cdot b = 0 \text{ or } a = 0,\, b \neq 0,\, b = 0$$

3 Homogeneous Functions and First-Order Homogeneous D.E

3.1 Homogeneous Functions

When you can't separate variables of a D.E, look at if it is Homogeneous.

First we examine the meaning of Homogeneous functions.

A function is said $f(x, y)$ is said to be homogeneous of degree "L" in x and y if the sum of the powers of x and y in each term of $f(x, y)$ is equal to "L".

i.e. if you have a function

$$\frac{x^2y + 2xy^2}{xy^2} = f(x, y)$$

Investigate if homogeneous → look at all terms.

First Order Homogeneous D.E

Check the sum of their powers and check if they are equal:

$$\left.x^2y = 3\,\text{deg}, \quad 2xy^2 = 3\,\text{deg}, \quad xy^2 = 3\,\text{deg}\right\}\text{Homogeneous of deg. 3}$$

Definition: A function $f(x, y)$ is said to be homogeneous of degree "m" in x and y if and only if (iff) this condition holds:

$$f(\alpha x, \alpha y) = \alpha^L f(x, y)$$

e.g. Determine the degree of each of the following functions; investigate if Homogeneous.

1. $f(x, y) = xy^4 - 4x^2y^3 - 3y^2x^3$
2. $f(x, y) = x^2 + y^2$
3. $f(x, y) = x^4y^4 + 2$

Solution

1. $f(x, y) = xy^4 - 4x^2y^3 - 3y^2x^3$

   Sub x for αx & y for αy:

$$\begin{aligned} f(\alpha x, \alpha y) &= (\alpha x)(\alpha y)^4 - 4(\alpha x)^2(\alpha y)^3 - 3(\alpha y)^2(\alpha x)^3 \\ &= (\alpha x)\alpha^4y^4 - 4(\alpha^2x^2)(\alpha^3y^3) - 3(\alpha^2y^2)(\alpha^3x^3) \\ &= \alpha^5xy^4 - 4\alpha^5x^2y^3 - 3\alpha^5x^3y^2 \\ &\therefore\ \alpha^5[x^1y^4 - 4x^2y^3 - 3x^3y^2] \end{aligned}$$

   $\therefore$ function is Homogeneous, with degree "$L = 5$"

$f(x, y) = 3x + 4y - 5$

Investigate if the function is homogeneous:

$$f(\alpha x, \alpha y) = \alpha f(x, y)$$

$$\begin{aligned} f(\alpha x, \alpha y) &= 3(\alpha x) + 4(\alpha y) - 5 \\ &= 3\alpha x + 4\alpha y - 5 \end{aligned}$$

Since α is not a common term for all three, it is **not homogeneous**.

$f(x, y) = 5x^2 + 4y^2 - 5y$

3.2 How to Test if a D.E is Homogeneous

How to test if a diff. eqn. is homogeneous, then solve with technique of solving non-homogeneous diff. eqn.

E.g. Determine if each of the following D.E is homogeneous, else give reason for answer:

1. $xy\dfrac{dy}{dx} - x^2 + ydx = 0$
2. $y(x^2+y^2)\dfrac{dy}{dx} =$
3. $xy\dfrac{dy}{dx} - x^2y^2\,dx = 0$
4. $x^2(x+y)dy - y^2$

$$\frac{dy}{dx} = \frac{M(x,y)}{N(x,y)} \qquad \text{(standard form of a first order Homogeneous diff. eqn)}$$

e) $xy\,dy - (x^2+y^2)dx = 0$

$$xy\,dy = (x^2+y^2)dx$$
$$\frac{dy}{dx} = \frac{x^2+y^2}{xy} \qquad \text{Step 1}$$

Ratio of sum-degree of numerator and denominator. To show:

$$M(x,y) = x^2+y^2, \qquad N(x,y) = xy$$
$$M(\alpha x, \alpha y) = \alpha^2 M(x,y)$$
$$M(\alpha x, \alpha y) = (\alpha x)^2 + (\alpha y)^2 = \alpha^2[x^2+y^2]$$

2) $(y^3x - 5x^2y^2)dx = (y^2+x^2y)dy$

$$\int (y^3x - 5x^2y^2) = \frac{dy}{dx}$$

$$\frac{}{y^3+x^2y} \qquad \textbf{Not homogeneous}$$

3.3 Procedure for Solving First Order Homogeneous D.E

When the functions are a product, $f(x,y)$ are separable.

SOLVING 1$^{\text{st}}$ Order HOMOGENEOUS D.E

Step 1 – Show whether the 1$^{\text{st}}$ order D.E is homogeneous, i.e. M and N are homogeneous function of the same degree.

Solve $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{xy}$ (Method of separation not possible, check if homogeneous.)

Step 2 – Express eqn (3) $\left(\dfrac{dy}{dx} = \dfrac{M(x,y)}{N(x,y)}\right)$ as $f(x,y) = F(v)$ where $v = \dfrac{y}{x}$ i.e. $\dfrac{y}{x}$ & $y = vx$.

Since D.E is homogeneous, solve with transformation.

Let $v = \dfrac{y}{x}$ and $y = vx$

Step 3 – Sub the $\dfrac{dy}{dx}$ of the L.H.S of eqn (3) by $v + x\dfrac{dv}{dx}$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

using product rule:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$\therefore\ v + x\frac{dv}{dx} = \frac{x^2 + (vx)^2}{x \times vx}$$

Step 4 – equate Step 2 and Step 3.

$$v + x\frac{dv}{dx} = F(v)$$

$$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{x^2v}$$

$$v + x\frac{dv}{dx} = \frac{x^2(1 + v^2)}{x^2v}$$

$$v + x\frac{dv}{dx} = \frac{1 + v^2}{v}$$

Homogeneous D.E is not separable because x & y are mixed, proceed to a maximum/minimum regrouping.

$\dfrac{dy}{dx}$ is separable, but if the functions are products, they are separable.

$$x\frac{dv}{dx} = \frac{1 + v^2}{v} - v = \frac{x(1 + v^2)}{v} - \frac{v^2}{1}$$

$$= \frac{x(1 + v^2) - v(v)}{v} \qquad \text{(check working)}$$

For a function $f(x, y)$ to be homogeneous, it must satisfy:

$$f(\alpha x, \alpha y) = \alpha^L f(x, y)$$

Standard form for a homogeneous D.E:

$$\frac{dy}{dx} = \frac{M(x, y)}{N(x, y)}$$

$$\frac{dy}{dx} = \frac{x + y}{x}$$

$$\left.v = \frac{y}{x}, \quad y = vx\right\}\ \text{USE}$$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{x + vx}{x}$$

$$v + x\frac{dv}{dx} = \frac{x(1 + v)}{x}$$

$$v + x\frac{dv}{dx} = 1 + v$$

$$x\frac{dv}{dx} = 1$$

$$x\frac{dv}{ddx} = \frac{1+v}{2}$$
$$\frac{2}{1+v}\,dv = \frac{1}{x}\,dx$$

Solving the second D.E:

$$x\frac{dv}{dx} = \frac{1+v}{2}$$
$$\int \frac{2}{1+v}\,dv = \int \frac{1}{x}\,dx$$
$$2\ln|1+v| = \ln|x| + c$$
$$\ln|1+v|^2 = \ln|x| + \ln c$$
$$\ln|1+v|^2 = \ln cx$$

Take exponential to cancel ln:

$$e^{\ln|1+v|^2} = e^{\ln cx}$$
$$|1+v|^2 = cx$$
$$\left(1+\frac{y}{x}\right)^2 = cx \qquad \text{General Soln.}$$

Who are noted of Separation of Variable:

$$v\,dv = \frac{1}{x}\,dx$$

Integrating both sides:

$$\int v\,dv = \int \frac{1}{x}\,dx$$
$$\frac{v^2}{2} = \ln x + C$$
$$v^2 = 2(\ln x + C)$$

recall $v = \frac{y}{x}$

$$\frac{y^2}{x^2} = 2(\ln x + C)$$
$$\frac{y}{x} = \sqrt{2(\ln x + C)}$$
$$y = x\sqrt{2(\ln x + C)}$$

or make y the subject of formula:

$$y = x\left(\frac{2\ln x}{x}\right)^{1/2}$$

b) Solve the D.E

$$\frac{dy}{dx} = \frac{x+3y}{2x}$$
$$v = \frac{y}{x},\ y = vx$$
$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$
$$v + x\frac{dv}{dx} = \frac{x+3(vx)}{2x} = \frac{x(1+3v)}{2x}$$

$$v + x\frac{dv}{dx} = \frac{1+3v}{2}$$
$$x\frac{dv}{dx} = \frac{1+3v}{2} - v = \frac{1+3v-2v}{2}$$
$$x\frac{dv}{dx} = \frac{1+v}{2}$$
$$\frac{2}{1+v}\,dv = \frac{1}{x}\,dx$$

Integrate both sides, values of $\ln|1+v|$, then find y.

4 D.E Reducible to Homogeneous Form (Non-Homogeneous)

4.1 Determinant of Coefficients Test

Case the Determinant of coefficients of x & y to have a non-zero value:

$$\begin{vmatrix} a & b \\ d & e \end{vmatrix} \neq 0$$

Case 2 – Determinant is zero:

$$\begin{vmatrix} a & b \\ d & e \end{vmatrix} = 0$$

Approaches used to convert Non-Homogeneous D.E to a Homogeneous D.E. (Reduction to first order)

For **Case 1**, we use the substitution procedure:

Let $x = X + h$, $\dfrac{dx}{dX} = 1$

$y = Y + k$, $\dfrac{dy}{dY} = 1$

$$\therefore \frac{dx}{dX} = \frac{dy}{dY}$$
$$\frac{dY}{dX} = \frac{dy}{dx}$$

Given: $\dfrac{dy}{dx} = \dfrac{a(X+h) + b(Y+k) + c}{d(X+h) + e(Y+k) + f}$

$$\frac{dY}{dX} = \frac{aX + ah + bY + bk + c}{dX + dh + eY + ek + f}$$
$$\frac{dY}{dX} = \frac{aX + bY + ah + bk + c}{dX + eY + dh + ek + f}$$
$$\frac{dY}{dX} = \frac{aX + bY + (ah + bk + c)}{dX + eY + (dh + ek + f)}$$

Solve for value of h and k by using simultaneous eqn. Then sub for h & k in original eqn.

$$ah + bk + c = 0 \quad \cdots (i)$$
$$dh + ek + F = 0 \quad \cdots (ii)$$

4.2 Worked Example

Solve the D.E $\quad \dfrac{dy}{dx} = \dfrac{x + 3y + 1}{3x - y - 7}$

$$\text{Investigate: } \begin{vmatrix} 1 & 3 \\ 3 & -1 \end{vmatrix} = -1 - 9 = -10 \neq 0$$

Step 1 (Transform)

Let $x = X + h$

$y = Y + k$

$$\frac{dy}{dx} = \frac{dY}{dX}$$

Step 2 (Substitute)

$$\frac{dY}{dX} = \frac{X + h + 3(Y + k) + 1}{3(X + h) - (Y + k) - 7}$$

Step 3 (Regroup to leave constant)

$$\frac{dY}{dX} = \frac{X + h + 3Y + 3k + 1}{3X + 3h - Y - k - 7}$$
$$\frac{dY}{dX} = \frac{X + 3Y + (h + 3k + 1)}{3X - Y + (3h - k - 7)}$$

Step 4 (Solve simultaneously)

$$h + 3k + 1 = 0$$
$$3h - k - 7 = 0$$

Multiply eqn 3 $\times 3$, & 11 by 1:

$$h + 3k = -1$$
$$3h - k = 7$$

Multiply $3 \times 3,\ 9k = -3$:

$$3h + 9k = -3$$
$$3h - k = 7$$
$$10k = -10$$
$$\therefore\ k = -1$$

Sub into eqn 1 to get h:

$$h + 3(-1) = -1$$
$$h - 3 = -1$$
$$h = -1 + 3$$
$$h = 2$$

Then solving the homogeneous d.e:

$$\frac{dY}{dX} = \frac{X + 3Y}{3X - Y}$$

(Transform)

Let $x = X - h \quad \therefore\ x - 2$

$$y = Y - k \quad \Rightarrow \quad y = Y - (-1) \quad \Rightarrow \quad y = Y + 1$$

$v = \dfrac{y}{x}, \quad y = vx$

$$\frac{dY}{dX} = \frac{v + X\,dv}{dX} \qquad \text{sub (ii) into eqn (3)}$$

$$\therefore\ \frac{dy}{dx} = \frac{X + 3(vx)}{X + 2(vx)}$$

$$= \frac{2x - vx}{X + 2vx}$$

Separating variables

$$\frac{X dv}{dX} = \frac{2 - v}{1 + 2v} - \frac{v}{1}$$

$$\frac{X dv}{dX} = \frac{2 - v - v(1 + 2v)}{1 + 2v}$$

$$= \frac{2 - v - v - 2v^2}{1 + 2v}$$

$$\frac{X dv}{dX} = \frac{2 - 2v - 2v^2}{1 + 2v}$$

Separate Variables

$$\int \frac{1 + 2v}{2 - 2v - 2v^2}\,dv = \int \frac{1}{X}\,dx$$

$$\int \frac{1 + 2v}{2(1 - v - v^2)}\,dv = \int \frac{1}{X}\,dx$$

$$-\frac{1}{2}\int \frac{1 + 2v}{-1 + v + v^2}\,dv = \int \frac{1}{X}\,dx$$

$$-\frac{1}{2}\ln(-1 + v + v^2) = \ln X + \ln K$$

(In of constant is a constant)

Multiply through by 2:

$$\ln(-1 + v + v^2) = -2(\ln X + \ln K)$$

$$\ln(-1 + v + v^2) = -2\ln XK \qquad (\log a + \log b = \log ab)$$

$$\ln(-1 + v + v^2) = \ln KX^{-2} \qquad (b\log a = \log a^b)$$

$$-1 + v + v^2 = KX^{-2}$$

$$-1 + v + v^2 = \frac{K}{X^2}$$

$$X^2(-1 + v + v^2) = K$$

$$X^2\left(-1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2\right) = K$$

$$-X^2 + X^2\frac{y}{X} + \frac{X^2y^2}{X^2} = K$$

$$-X^2 + XY + Y^2 = K$$

recall $x = X + h$, $y = Y - k$

$$\therefore\ X = x - 2, \qquad y = Y - 2$$

4.3 Worked Example: Case 2 (Determinant = 0)

2) Solve the Reduce the D.E to a first order Homogeneous:

$$\frac{dy}{dx} = \frac{3x + 4y - 2}{3x - 4y - 3}$$

$$\begin{vmatrix} 3 & -4 \\ 3 & -4 \end{vmatrix} = -12 - (-12) = -12 + 12 = 0$$

$\therefore$ Using **Case 2**.

Using a diff. transformation. Let $X =$ with Num. or den.

Let $X = 3x - 4y$

$$\frac{dX}{dx} = 3 - 4\frac{dy}{dx}$$

$$\frac{dX}{dx} - 3 = -4\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{dX}{dx} \cdot 3 \div -4$$

$$= \left(\frac{dX}{dx} - 3\right) \div -4$$

$$\therefore\ -\frac{1}{4}\frac{dX}{dx} + \frac{3}{4} = \frac{dy}{dx}$$

Since it should be, since it only has 1 constant, sub for both sides of the D.E:

$$-\frac{1}{4}\frac{dX}{dx} + \frac{3}{4} = \frac{X-2}{X-3}$$

$$-\frac{1}{4}\frac{dX}{dx} = \frac{X-2}{X-3} - \frac{3}{4}$$

$$-\frac{1}{4}\frac{dX}{dx} = \frac{4X - 8 - 3(X-3)}{(X-3)4}$$

$$-\frac{1}{4}\frac{dX}{dx} = \frac{4X - 8 - 3X + 9}{4X - 12}$$

$$-\frac{1}{4}\frac{dX}{dx} = \frac{X+1}{4X-12}$$

Separate: multiply through by 4

$$-\frac{dX}{dx} = \frac{4(X+1)}{4(X-3)}$$

$$-\frac{dX}{dx} = \frac{X+1}{X-3}$$

$$\int \frac{X-3}{X+1}\, dX = \int -dx$$

Note: Polynomial division; use long division if both degrees are equal.

$$\begin{array}{r} 1 \\ \hline X+1 \,\big)\, X-3 \\ -(X+1) \\ \hline -X-1 \\ \hline -4 \end{array} \qquad \int -4/(X+1)$$

$$\int \left(1 - \frac{4}{X+1}\right) dX = \int -dx$$
$$X - 4\ln|X+1| = -x + K$$

5 Second Order Linear D.E with Constant Coefficients

Second order linear D.E, presented as O.D.E.

Given: Highest order = 2, (2nd order)

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = G(x) \quad \cdots (i)$$

where $r = \dfrac{R(x)}{Q(x)}$

$$y = \frac{1}{P(x)}\left[\int P(x)Q(x)\,dt + c\right]$$

$1.f = P(x) = e^{\int P(x)\,dx}$

Sub in

Bernoulli:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \ \text{ (non-linear)}$$
$$a \cdot b = 0 \text{ if } a = 0,\ b \neq 0, \ \text{ or } b = 0$$

Note: exponential of $B_3(t)$ values cannot be 0.

5.1 From eqn (i)

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

This is a second order O.D.E.

$Q(x)$ and $R(x)$ are all constant coefficients, or can represent functions of x or constant coe.

$$1) \text{ If } F(x) = 0, \text{ we have a second order homogeneous D.E} \quad \text{(This is)} \quad \cdots (11)$$

Homogeneous second order – Considering the P, Q and R as all constant coefficients, or can represent constant with a, b, c respectively:

$$ay'' + by' + cy = 0 \quad \cdots (111)$$

$$2) \text{ If } F(x) \neq 0, \text{ we have a second order} \quad \cdots (1-n)$$
$$y = \frac{1}{P(x)}\left[\int (1-n)P(x)Q(x)\,dx + c\right]$$

Nonhomogeneous D.E (they do not contain x)

$$1.f = P(x) = e\int (1-n)P(x)\,dx$$

Second order linear D.E is going to have a second order.

∴ ay”+by’+cy=0 is the auxiliary equation (characteristics equation).

So $1.f = am^2 + bm + c = 0$

Solving our tried Soln to the D.E (Gives multiple exponential Soln):

$$y' = me^{mx}, \qquad y'' = m^2 e^{mx}$$

$$\text{Assuming our tried Soln as } y_1' = me^{mx}, \quad y_2'' = m^2\, e^{mx}$$

Then with a, b, c constant coefficients, or can represent (this do not contain x)

If $P(x) = 0$, then the D.E is:

$$ay'' + by' + cy = 0$$

$$\text{1) If } D > 0, \text{ root is going to be}$$

$$m_1 = m_2$$

$$\text{if } D = 0$$

$$\text{root is real and}$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

Case 3: Unreal and Complex (i)

$$\text{if } D < 0, \text{ root is going to be}$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

if D: $D = b^2 - 4ac$

5.2 Solving Homogeneous Second Order D.E: Cases

Case 1: If $D > 0$, root is going to be real, and distinct

$$m_1 = m_2$$

Then the sol^{n} to the D.E=

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

Case 2: real and equal roots

If $D = 0$, then the sol^{n} to the D.E is

$$y = C_1 e^{mx} + C_2 x e^{mx}$$

Case 3: Unreal and Complex roots

Roots of the characteristics eqn.

$$(D > 2, \; D = 0, \; D < 0)$$

5.3 Worked Examples

Solve $y'' - 5y' + 6y = 0$

$$m^2 - 5m + 6 = 0$$
$$(m-2)(m-3) = 0$$
$$m = 2,\ m = 3$$
$$y = C_1e^{2x} + C_2e^{3x}$$

Solve $y'' + y' - 6y = 0$

Auxiliary/characteristics eqn.

$$m^2 + m - 6 = 0$$
$$(m^2 - 3m) + (2m + 6) = 0$$
$$m(m-3) + 2(m-3) = 0$$
$$(m-3)(m+2) = 0$$
$$m = 3,\quad m = -2$$
$$y = C_1e^{3x} + C_2e^{-2x}$$

5.4 Solving Homogeneous D.E with Given Conditions

Given $G(x) = 0$ (Homogeneous), $y(0) = 1$ & $y'(0) = 0$

Solve the D.E $y'' + y' - 6y = 0$ and $y(0) = 1$, $y'(0) = 0$

$$m^2 + m - 6 = 0$$
$$(m^2 + 3m) - (2m - 6) = 0$$
$$m(m+3) - 2(m+3) = 0$$
$$m(m+3) - 2(m+3) = 0 \text{ G.S}$$
$$m = -3,\quad m = +2$$
$$y = C_1e^{-3x} + C_2e^{2x}$$
$$y' = -3C_1e^{-3x} + 2C_2e^{2x}$$
$$\text{I.V.P}$$
$$1 = C_1e^{-3(0)} + C_2e^{2(0)}$$
$$1 = C_1(1) + C_2(1)$$
$$1 = C_1 + C_2 \quad \cdots (1)$$

Sub eqn (1) into (11) $\Rightarrow$

$$3 - 8C_2 = 2C_2$$
$$3(1 - C_2) = 2C_2$$
$$3 - 3C_2 = 2C_2$$
$$3 = 5C_2$$
$$C_2 = \frac{3}{5}$$
$$C_1 = 1 - C_2 = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\text{G.S.:} \quad y = C_1e^{3x} + C_2e^{2(0)}$$
$$1 = C_1e^{-3(0)} + C_2e^{2(0)}$$

Plug (IV) into (3):

$$2(-1) - (-2) - 0(x) - (-1)x^2 = x^2$$
$$-2 + 2 - 0 + x^2 = x^2$$
$$\therefore \; x^2 = x^2$$

5.5 Non-Homogeneous Second Order D.E (Particular Solution)

Recall $y'' - y = x^2$. Since $x^2 \neq 0$, it is Non-Homogeneous.

$$y = y_h + y_p$$

To find y_h:

$$y'' - y = 0$$
$$m^2 - 1 = 0$$
$$m^2 = 1 \qquad \therefore \; m = \pm\sqrt{1}$$
$$m = 1 \text{ or } m = -1$$

Since roots are real & distinct:

$$y = C_1e^{x} + C_2e^{-x} \quad (\text{keep } y_h)$$

To find y_p, since x^2 $(G(x))$ is a quadratic, replace $y_p(x)$ with:

$$Ax^2 + Bx + C$$

$$y = y_h + y_p \quad \cdots (1)$$
$$y = Ax^2 + Bx + C$$
$$\left.\begin{aligned} y' &= 2Ax + B \\ \\ y'' &= 2A \end{aligned}\right\} 2$$

Recall $y'' - y = x^2$

Sub eqn (2) into (1):

$$2A - (Ax^2 + Bx + C) = x^2$$
$$2A - Ax^2 - Bx - C = x^2$$
$$(2A - C) - Bx - Ax^2 = x^2 \quad \cdots (11)$$

Equating Coefficients:

$$2A - C = 0$$
$$-Bx = 0 \Rightarrow -Ax^2 = x^2$$
$$-A = 1 \Rightarrow A = -1$$
$$B = 0$$

To find C:

$$2A - C = 0$$
$$2(-1) - C = 0$$
$$-2 - C = 0$$
$$-C = 2$$
$$C = -2$$

$$A = -1,\ B = 0,\ C = -2 \quad \cdots (iv)$$

Plug (iv) into (3):

$$y_p = Ax^2 + Bx + C = -x^2 + 0(x) - 2$$
$$y_p = -x^2 - 2$$

The G.S:

$$y = y_h + y_p = C_1 e^x + C_2 e^{-x} - x^2 - 2$$

5.6 Non-Homogeneous D.E with Initial Conditions

When $G(x) \neq 0$ (nonhomogeneous)

Solve $y'' - 6y' + 9y = 1, \quad y(0) = 0$ and $y'(0) = 0$

$$m^2 - 6m - 9 = 0$$
$$(m+3)(2m-6) = 0$$
$$(m+3)(m-3) = 0$$
$$m + 3 = 0, \quad m(m+3) - 2(m+3) = 0$$
$$m(m+3) - 2(m+3) \Rightarrow G.S$$
$$m = -3,\ m = +2$$

5.7 Exact D.E Reducible Form – Note

$$\frac{dy}{dx} = ax + bg + C \qquad \cdots (1)$$
$$dx + ey + f$$

where a, b, ε, d and e are constant coe. of the variable of the x & y while C & F are constants, but not coe. of any variable. At this level, the D.E is not homogeneous, bcos of the presence of C & F which are not coe. of the variable x & y.

It should be noted that at least one of C & F has a non-zero value, otherwise the D.E could have been homogeneous. However, we can make eqn (1) Homogeneous if

6 Exact Differential Equations

If a D.E is not separable variable, not homogenous, not first order linear, otherwise we test if its "**exact**".

$$\frac{dy}{dx} + D(x)y = Q(x)$$

Example. The D.E $M(x, y)\,dx + N(x, y)\,dy = 0$ is the standard form, and its exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

condition holds, and the sol$^{\text{n}}$ is the integral of $M\,dx$ + the integral of the terms in N without x, $dy = C$

i.e.

$$\int M\,dx + \int (\text{terms in } N \text{ without } x)\,dy = C$$

Test for exactness

$$\int x - 3(dx) = \int -dx$$

Note: Polynomial division, use long division if both degrees are equal.

6.1 Worked Example

Given $M(x, y) = x^2 + y^2$, $N(x, y) = xy$

$$\begin{aligned} M(\alpha x, \alpha y) &= \alpha^2 m(x, y) \\ M(\alpha x, \alpha y) &= (\alpha x)^2 + (\alpha y)^2 \\ &= \alpha^2[x^2 + y^2] \end{aligned}$$

Exact D.E Example

Solve the D.E:

$$(3x^2y^2 + 4x)\,dx + (2x^3y + 3y^2 + 5y)\,dy = 0, \quad y(0) = 1$$

Test: Take Partial derivatives

$$\begin{aligned} M &= (3x^2y^2 + 4x) \\ N(x, y) &= (2x^3y + 3y^2 + 5y) \end{aligned}$$

$$\begin{aligned} \frac{\partial M}{\partial y} &= 6x^2y \\ \frac{\partial N}{\partial x} &= 6x^2y \end{aligned}$$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} = 6x^2y$, therefore the D.E is exact.

Find Solution

$$\int (3x^2y^2 + 4x)\,dx + \int (3y^2 + 5y)\,dy = C$$

$$x^3y^2 + 4\frac{x^2}{2} + \frac{3y^3}{3} + \frac{5y^2}{2} = C$$

$$x^3y^2 + 2x^2 + y^3 + \frac{5}{2}y^2 = C \qquad \text{General Solution}$$

Sub for $y(0) = 1$:

$$0^3(1)^2 + 2(0)^2 + 1^3 + \frac{5(1)^2}{2} = C$$

$$1 + \frac{5}{2} = C$$

$$\frac{2+5}{2} = C$$

$$C = \frac{7}{2} = 3.5$$

Sub for C:

$$x^3y^2 + 2x^2 + y^3 + \frac{5}{2}y^2 = \frac{7}{2} \qquad \text{Particular Solution}$$

6.2 Second Example

2) Solve the D.E

$$\frac{dy}{dx} = \frac{2xy - \sin x}{-(x^2 - \cos y)}$$

Cross multiply to get standard form:

$$(2xy - \sin x)\,dx = -(x^2 - \cos y)\,dy$$

$$(2xy - \sin x)\,dx + (x^2 - \cos y)\,dy = 0$$

$$M(x,y) = 2xy - \sin x$$

$$N(x,y) = x^2 - \cos y$$

$$\frac{\partial M}{\partial y} = 2x - 0 = 2x$$

$$\frac{\partial N}{\partial x} = 2x, \quad -x^2 \to 2x - 0$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2x$, D.E is exact.

$$\int (2xy - \sin x)\,dx + \int (-\cos y)\,dy = C$$

$$= \frac{2x^2y}{2} + (-\cos x) + \sin y = C$$

$$x^2y + \cos x + \sin y = C$$

7 Linear First Order D.E

A D.E of the form $\dfrac{dy}{dx} + P(x,y) = Q(x)$ is a linear first order D.E where $P(x)$ and $Q(x)$ are continuous functions of x; you confirm that solution to the D.E, using an integrating factor, method of variation of constant.

Standard form:

$$\frac{dy}{dx} + P(x,y) = Q(x)$$

Solve $\quad xy' - y = x^2 \cos x, \quad y(\pi/2) = \pi$

$$x\frac{dy}{dx}$$

Standard Form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

divide through by x:

$$x\frac{dy}{dx} - \frac{1}{x}y = \frac{x^2 \cos x}{x}$$
$$\frac{dy}{dx} - \frac{1}{x}y = x\cos x$$

Step 2: Find Integrating Factor

$$P(x) = -\frac{1}{x}$$
$$Q(x) = x\cos x$$
$$1.f = B(x) = e^{\int P(x)\,dx}$$
$$= e^{\int -1/x\,dx}$$
$$= e^{-\ln x}$$
$$B(x) = e^{\ln x^{-1}}$$
$$= x^{-1} = \frac{1}{x}$$

Step 3

General Soln

$$y = \frac{1}{Bx}\left[\int Bx\,Q(x)\,dx + C\right]$$
$$y = \frac{1}{1/x}\left[\int \frac{1}{x}x\cos x\,dx + C\right]$$
$$= x\left[\int \cos x\,dx + C\right]$$
$$= x\left[\sin x + C\right]$$
$$= x\sin x + Cx \qquad \text{Gen. Soln}$$

To get the particular solution:

Factor from $y = x\sin x + Cx$

$$y = \frac{\pi}{2}\sin$$

$$\pi = \frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) + C\left(\frac{\pi}{2}\right)$$
$$\pi = \frac{\pi}{2}\sin(180/2) + \frac{\pi}{2}C$$
$$\pi = \frac{\pi}{2}\sin 90 + \frac{\pi}{2}C$$
$$\pi = \frac{\pi}{2} + C\frac{\pi}{2}$$
$$2\pi = \pi + C\pi$$
$$2\pi - \pi = C\pi$$
$$\pi = C\pi \quad \therefore\ C = 1$$

In terms of angle, $\pi = 180^\circ, \quad 2\pi = 360, \quad \pi/2 = 90$

$\therefore$ Particular Soln $= x\sin x + x$

8 Bernoulli's Equations

8.1 Definition

The non-linear D.E of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \cdots (*)$$

$\Big\{$Linear if derivatives are not multiplying function$\Big\}$

Power of function must be 1

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \cdots (*)$$

nonlinear, where n is a real number, not equal to 0 (if $n = 0$, y removed) or 1 is called a Bernoulli d.e.

Note 1: If $n = 0$, the Bernoulli equation in (*) reduces to a linear d.e. of the form:

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \cdots (11)$$

Note 2: If $n = 1$, eqn (*) becomes:

$$\frac{dy}{dx} + P(x)y = Q(x)y$$

$$\frac{dy}{dx} + P(x)y - Q(x)y = 0$$

$$\frac{dy}{dx} + y(P(x) - Q(x)) = 0 \quad \to (111)$$

$\hookrightarrow$ Solved using Separation of variable, which is linear, and can be solved by variable separation.

If $n \neq 0$ and $n \neq 1$: non-linear eqn in 1, can be solved by reducing the d.e to a linear first order using appropriate approach to your soln.

The Approach

8.2 Worked Example 1

Solve the D.E

$$x\frac{dy}{dx} + y = y^2, \qquad y(1) = 2$$

Divide through by 1:

$$\frac{x}{x}\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x}y^2$$

$$\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}y^2$$

$P(x) = 1/x, \;\; Q(x) = 1/x, \quad n = 2$

Let Integrating factor:

$$1.F = B(x) = e^{\int (1-n)P(x)\,dx}$$

$$= B(x) = e^{\int (1-2)1/x\,dx}$$

$$= e^{\int (-1)1/x\,dx}$$

$$= e^{-\int 1/x\, dx}$$
$$= e^{-\ln x}$$
$$= e^{\ln x^{-1}}$$
$$B(x) = x^{-1} = 1/x$$

Step 3: The Solution

$$y^{1-n} = \frac{1}{B(x)}\left[\int (1-n)B(x)Q(x)\, dx + C\right]$$
$$y^{1-2} = \frac{1}{1/x}\left[\int (1-2)\,\frac{1}{x} \times \frac{1}{x}\, dx + C\right]$$
$$y^{-1} = x\left[\int (-1)\frac{1}{x^2}\, dx + C\right]$$
$$y^{-1} = -x\left[\int \frac{1}{x^2}\, dx + C\right]$$
$$y^{-1} = -x\left[\int (x^{-2})\, dx + C\right]$$
$$y^{-1} = -x\left[\frac{x^{-2+1}}{-2+1} + C\right]$$
$$y^{-1} = -x\left[\frac{x^{-1}}{-1} + C\right]$$
$$y^{-1} = -x\left[-x^{-1} + C\right]$$
$$y^{-1} = x \cdot x^{-1} - Cx$$
$$y^{-1} = x \cdot \frac{1}{x} - Cx$$
$$y^{-1} = 1 - Cx$$
$$\frac{1}{y} = 1 - Cx$$
$$y(1 - Cx) = 1$$
$$y = \frac{1}{1 - Cx} \qquad \text{G.S}$$

Apply I.C: $y(1) = 2$

$$2 = (1 - C(1))^{-1} \qquad \text{— G.S}$$
$$2 = \frac{1}{1 - C(1)}$$
$$2(1 - C(1)) = 1$$
$$2 - 2C = 1$$
$$-2C = 1 - 2$$
$$2C = 1$$
$$C = 1/2$$

$$\therefore\ P.S: \quad y = \left(1 - \frac{1}{2}x\right)^{-1} = \left(1 - \frac{x}{2}\right)^{-1}$$
$$y = \frac{1}{1 - x/2}$$

8.3 Worked Example 2

Solve the D.E:

$$\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}y^2 \qquad \text{[Note 2: If } n = 1\text{, eqn (1) becomes]}$$

$$P(x) = 1/x, \quad Q(x) = 1/x, \quad n = 2$$

Recall separable variable D.E:

$$\frac{dy}{dx} = f(x) \cdot g(y) \qquad \text{Separable}$$

$$\frac{dy}{dx} + P(x)y = Q(x) \qquad \text{first order linear d.e}$$

9 Additional Worked Homogeneous D.E Example

Solve $\dfrac{dy}{dx} = \dfrac{x+3y}{3x}$

$$v = \frac{y}{x}, \qquad y = vx$$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$\therefore\ v + x\frac{dv}{dx} = \frac{x+3(vx)}{3x}$$

$$= \frac{x(1+3v)}{3x}$$

$$v + x\frac{dv}{dx} = \frac{1+3v}{3}$$

$$x\frac{dv}{dx} = \frac{1+3v}{3} - v$$

$$x\frac{dv}{dx} = \frac{1+3v-3v}{3}$$

$$\frac{x\,dv}{9_x} = \frac{1+3v}{2}$$

Methods of Separation of variable

$$v\,dv = \frac{1}{x}(dx)$$

Integrating both sides:

$$\int v\,dv = \int \frac{1}{x}\,dx$$

$$\frac{v^2}{2} = \ln x + C$$

$$v^2 = 2(\ln x + C)$$

recall $v = \dfrac{y}{x}$

$$\frac{y}{x} = \sqrt{2(\ln x + C)}$$

$$y = x\sqrt{2(\ln x + C)}$$